Zero-point flux fluctuations switch a quantum-dot Josephson loop from 0 to π

A superconducting loop closed by a quantum-dot Josephson junction and sharing its inductance with a microwave mode is a π loop whenever the dot is in its doublet state. We solve the full electron–photon problem of this circuit, an interacting Anderson impurity in a BCS ring threaded by the quantized flux of a lumped LC mode, by DMRG, with the Peierls vertex $e^{ig(a+a^\dagger)}$ kept to all orders. Three results follow. (i) The zero-point flux of the mode moves the singlet–doublet (0–π) boundary of the junction: at a mode impedance $Z \simeq 2\,\mathrm{k\Omega}$ ($g \simeq 0.5$) a junction that is a 0 junction in isolation becomes a π junction, by an amount that crosses over from the adiabatic flux average at $\hbar\omega \ll E_A$ to the Franck–Condon reduction $t_J \to t_J e^{-g^2/2}$ of the tunnelling when $\hbar\omega$ exceeds the Andreev scale. (ii) Deep in the adiabatic regime the exact solution reduces, to $10^{-5}\Delta$ in energy and 0.2% in current, to an rf-SQUID (fluxonium) Hamiltonian with the microscopic $E(\varphi)$ of the junction: the "photon condensate" with $\langle a\rangle \neq 0$ is the classical screening instability $\beta = 2\pi L I_c/\Phi_0 > 1$, its ground state is the cat state of the two persistent-current directions, and the threshold is pushed above $\beta = 1$ by the zero-point motion of the flux. (iii) The electron×photon product state used in recent mean-field treatments of this circuit orders where the exact solution does not, both for $E_J \lesssim 3\hbar\omega$ and at the 0–π boundary, where a static moment replaces the Kramers doublet. We give the parameters, in Al and NbTiN, at which the flux-driven parity switch and the cat state are observable in two-tone spectroscopy.

A superconducting loop closed by a π junction carries a spontaneous persistent current when its inductance exceeds $\Phi_0/2\pi I_c$ [1–4], and with a small capacitance the two current directions form a flux qubit that needs no external field [5–10]. The same circuit is the counterexample to the no-go theorem for superradiance: a mode that couples through its magnetic flux can condense a static field $\langle a\rangle \neq 0$ [11–18], the condition being the magnetostatic instability of the matter [17, 19]. In all of these descriptions the junction is a fixed function $E(\varphi)$, or electrons and mode are decoupled in a product state [20]. A quantum-dot Josephson junction is not a fixed $E(\varphi)$: it is a π junction in its doublet state [21–28], its singlet–doublet transition is read out through the frequency of a transmon or resonator [29–31], shunted by a superinductor it is a protected spin qubit [32, 33], and whether it is 0 or π is decided by U, the level width Γ and Δ on a scale of a few GHz, comparable to the mode frequency. Here we treat the electrons of the junction and the quantized flux on the same footing and solve the electron–photon problem exactly by DMRG, as done for normal chains [34, 35] and ladders [36] in a cavity. The zero-point flux of the mode turns out to be a control parameter of the 0–π transition; deep in the adiabatic regime the loop is exactly a fluxonium with the microscopic $E(\varphi)$; and the product-state mean field fails where $E_J \lesssim \hbar\omega$ or $U \simeq U_c$.

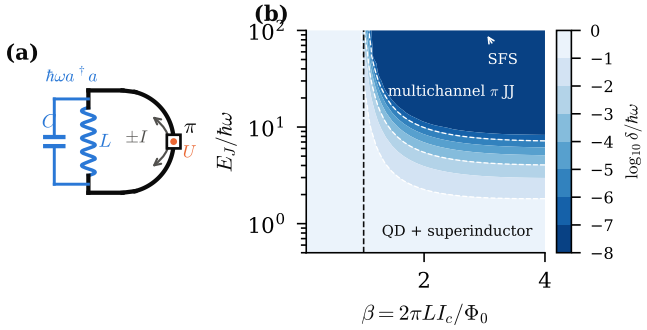


FIG. 1. (a) A superconducting loop closed by an interacting quantum dot (charging energy U) shares its inductance L with an LC mode. In the doublet state of the dot the loop is a π loop and its two degenerate states carry opposite persistent currents $\pm I$. (b) Tunnel splitting δ between the two current states in the plane of the screening parameter $\beta = 2\pi L I_c/\Phi_0$ and $E_J/\hbar\omega$ [Eq. (2) with the DMRG $E(\varphi)$ of the $U = 2\Delta$ junction]. White lines: $\delta/\hbar\omega = 10^{-6}, 10^{-3}, 10^{-1}$; dashed: $\beta = 1$.

Model. The circuit of Fig. 1(a) is an s-wave BCS chain of $L-1$ sites (hopping t, pairing Δ) closed by an Anderson impurity (site 0; $\epsilon_d = -U/2$, repulsion U, tunnelling t_J to both ends) and threaded by the flux of one lumped LC mode:

$$
\begin{aligned}
H = {} & \hbar\omega\, a^\dagger a - t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.} \right) \\
& + \Delta \sum_{r=1}^{L-1} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.} \right) + \epsilon_d\, n_0 + U n_{0\uparrow} n_{0\downarrow} \\
& - t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.} \right) \\
& - t_J \sum_{\sigma} \left(e^{i\varphi_{\mathrm{cl}}}\, e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.} \right) . \qquad (1)
\end{aligned}
$$

The mode enters through the displacement operator $D = e^{ig(a+a^\dagger)}$, so diamagnetic terms are kept to all orders [37, 38]; $g = 2\pi\Phi_{\mathrm{zpf}}/(h/e)$ is the single-electron phase of the zero-point flux, $\varphi = \pi$ is one flux quantum

$\Phi_0 = h/2e$, and for a mode of impedance Z sharing its inductance with the loop, $g^2 = \pi Z/4R_Q$ ($R_Q = h/4e^2$). $\varphi_{\rm cl}$ is a static flux, used as the $\Phi_0/2$ bias of a weak link or as an infinitesimal seed. Placing the whole phase on the dot bond is a statement about the circuit, not a gauge choice: a static phase can be spread along the ring by a gauge transformation with the pairing transformed covariantly [38] (Supplemental Material, SM), but the quantum part cannot [39], since moving a fraction λ of gX onto other bonds generates $\hbar\omega(\lambda NP + \lambda^2 N^2)$, a capacitive coupling of the displaced charge N to the charge P of the mode. Eq. (1) is the circuit in which the voltage of the mode drops across the junction, the only high-impedance element of the loop; the omitted terms are of relative order $\hbar\omega g^2/E_A$. Unless stated otherwise, $\Delta = t$ (energies in Δ, currents in $e\Delta/\hbar$), $t_J = 0.5\Delta$ ($\Gamma = 2t_J^2/t = \Delta/2$), $L = 10 > \xi \simeq t/\Delta$ (finite-size and h/e corrections [40] are $\le 2\%$, SM), $\varphi_{\rm cl} = 0$.

Fermion parity s is conserved, so the ground-state energy of the ring at fixed phase, $E_s(\varphi) = \sum_m E_m^{(s)} \cos m\varphi$, is defined sector by sector: the singlet (even) sector has $E_2 < 0$ (a 0 junction), the doublet (odd) sector $E_2 > 0$ (a π junction), and $E_s(\varphi) = E_s(-\varphi)$ exactly because $H[\varphi_{\rm cl} = 0]$ is real. The persistent current is $I = -(e/\hbar)\partial H/\partial\varphi_{\rm cl}$, the particle current through the dot bond dressed by D, and $\langle[H, P]\rangle = 0$ gives the sum rule $\langle I\rangle = (e/\hbar)(\hbar\omega/2g)\langle a + a^\dagger\rangle$ in every eigenstate: the current *is* the static displacement of the mode, and the difference of the two sides is our acceptance test for every DMRG state.

Two limits. If $\hbar\omega$ is small compared with the excitation gap E_A of the ring at fixed phase and parity, the electrons follow the flux adiabatically and, in each sector,

$$H_{\rm BO}^{(s)} = \frac{\hbar\omega}{4}\left(P^2 + X^2\right) + E_s(\varphi_{\rm cl} + gX), \qquad [X, P] = 2i, \tag{2}$$

which in the Cooper-pair phase $\theta = 2gX$ is $4E_C n^2 + \frac{1}{2}E_L\theta^2 + E_s(\theta/2)$ with $E_L = \hbar\omega/8g^2$, $E_C = \hbar\omega g^2$: the rf-SQUID of Refs. [8, 41], a fluxonium [42] whose Josephson element is the dot. A static displacement appears classically when the paramagnetic response of a π loop exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2|E''(0)|}{\hbar\omega} = \frac{8g^2E_J}{\hbar\omega} = \frac{2\pi LI_c}{\Phi_0} > 1, \tag{3}$$

the screening condition of Refs. [1, 2], the superradiance condition of the circuit of Bamba *et al.* [13] and the single-loop form of the magnetostatic criterion of Refs. [17–19]. Zero-point fluctuations oppose the condensation [43]: at finite $E_J/\hbar\omega$ the ground state is the symmetric superposition of the two current states, split by $\delta \sim \hbar\omega e^{-S}$ with $S \propto E_J/\hbar\omega$, and a sharp transition exists only for $E_J/\hbar\omega \to \infty$ at fixed β [44]. In the opposite limit, $\hbar\omega \gg E_A$, the mode averages the Peierls factor over its vacuum: a Lang–Firsov transformation [45] gives $H_{\rm ring}$ with $t_J \to t_J e^{-g^2/2}$ plus $O(t_J^2/\hbar\omega)$, i.e. the zero-point flux reduces the level width of the dot by the Franck–Condon factor e^{-g^2}. Neither limit is assumed below.

Methods. We solve Eq. (1) by two-site DMRG (TeNPy [46]) on the full electron–photon space: the ring in folded order next to a bosonic site (Fock cutoff 20–62, last Fock weight $< 10^{-7}$), parity and S^z conserved, $\chi = 200$–400; benchmarks against BdG (10^{-14}) and against exact diagonalization of a four-site ring with a photon, sum rule included (10^{-7}). Seeded scans use two initial states (vacuum side, polarized side) and seed annealing, because the two current states are split only by δ and by $2I^\star\varphi_{\rm cl} \sim 10^{-5}\Delta$ (SM). We compare with (i) Eq. (2) diagonalized exactly with the DMRG $E_s(\varphi)$ ("circuit model"), (ii) the Lang–Firsov ring, and (iii) the self-consistent electron×photon product state (Hartree–Fock BdG electrons times a photon state, "mean field", MF), the ansatz of Refs. [17, 20] for this circuit.

Flux-driven 0–π switch. Figure 2 contains the first result. Without the mode the junction is a 0 junction for $U < U_c^0 = 0.94\Delta$, where the singlet is the ground state at zero flux, and a π junction above (SM). With the mode we compute the ground state of Eq. (1) in both parity sectors at $\varphi_{\rm cl} = 0$, without any seed, for $\hbar\omega = 0.01$–$1\,\Delta$ and $g \le 1$, and locate the boundary $U_c(g, \omega)$ of the *loop* from the crossing $E_{\rm odd} = E_{\rm even}$. At $U = 0.9\Delta$ and $\hbar\omega = 0.1\Delta$ [Fig. 2(a)] the doublet–singlet energy difference falls from $+0.010\Delta$ at $g = 0$ to -0.008Δ at $g = 0.4$ and -0.065Δ at $g = 1$: the loop switches from 0 to π at $g \simeq 0.28$, a mode impedance $Z \simeq 0.6\,\mathrm{k\Omega}$, with nothing but the vacuum of the mode acting on the junction. The boundary $U_c(g, \omega)$ [Fig. 2(b)] moves down at every ω: to 0.83Δ at $g = 0.5$ and to 0.65, 0.60 and 0.57Δ at $g = 1$ for $\hbar\omega = 0.1$, 0.3 and $1\,\Delta$, i.e. by 30–40% of U_c^0.

The mechanism changes with $\hbar\omega$. For $\hbar\omega \ll E_A$ the exact boundary coincides with that of the circuit model (2) (dashed; 0.889 against 0.889Δ at $g = 0.4$ and 0.813 against 0.815Δ at $g = 0.8$ for $\hbar\omega = 0.01\Delta$): the zero-point motion of the flux samples $E_s(gX)$ around $\varphi = 0$, where the singlet surface curves up ($E''_{\rm even}(0) = +0.20\Delta$) and the doublet surface curves down, so that the singlet loses and the doublet gains zero-point energy, $\delta E_s \simeq \frac{1}{2}g^2E_s''(0)\langle X^2\rangle$, and for gX of order one the doublet lowers its energy further by letting the flux relax towards $\Phi_0/2$, where its minimum lies. This is the quantum version of the parity switch by a free flux discussed below. As $\hbar\omega$ grows the exact boundary departs from the circuit model towards the Lang–Firsov line (dotted), by 0.015, 0.032 and 0.040Δ at $g = 0.8$ for $\hbar\omega = 0.1$, 0.3 and $1\,\Delta$: the mode no longer follows the junction but reduces its level width to Γe^{-g^2}, and the singlet, which is bound by Γ, loses against the doublet. The adiabatic shift is bounded by the condensation energy of the π well, the Franck–Condon shift by Γ alone; at $\hbar\omega = \Delta$ the corrections $O(t_J^2/\hbar\omega)$ still keep the exact boundary well above the Lang–Firsov line. The dressing of the junction is visi-

ble in the Peierls factor and in the photon–electron entanglement [Fig. 2(c)]: $|\langle D\rangle|$ follows $e^{-g^2/2}$ for $\hbar\omega \gtrsim 0.3\Delta$, where the mode is only squeezed by its vacuum, and falls faster in the adiabatic regime, where it is displaced towards the π well; the entanglement entropy is largest there (0.26 at $g = 0.8$, $\hbar\omega = 0.01\Delta$), because the flux state depends on the electronic sector.

The effect is not an artefact of the one-bond vertex: the neglected capacitive terms are of relative order $\hbar\omega g^2/E_A$, and the shift of U_c at $\hbar\omega = 0.01\Delta$, where they are negligible, is already half of the one at $\hbar\omega = 0.3\Delta$. Nor is it a mean-field effect: the Hartree–Fock product state replaces the Kramers doublet by a static moment and places the boundary at 0.82Δ for every g (next section). The prediction is that a dot junction gate tuned close to its singlet–doublet boundary, in a loop whose inductance is a superinductor of impedance $Z \gtrsim 1\,\mathrm{k\Omega}$, is driven into the π phase by the impedance alone, with the switch visible as the jump of the resonator frequency that marks the parity transition in Refs. [29, 31].

Persistent-current phase of the π loop. We now take $\hbar\omega = E_J(U = 2\Delta)/10 = 7.39\times10^{-4}\Delta \ll E_A$, so that the classical line $\beta = 1$ sweeps across $g \le 0.4$ ($Z \le 1.3\,\mathrm{k\Omega}$) as U runs from U_c to 8Δ while $E_J/\hbar\omega$ falls from 15 to 2, and compute the ground state on a grid of 427 points ($g = 0.02$–0.40, 14 values of U, both sectors for $U \le \Delta$, seed 10^{-3}, $\chi = 300$). Figure 3 shows the seeded persistent current of the full DMRG (a), the seed-free order parameter of the circuit model, the tunnel splitting $\delta(U, g)$ between the two current directions (b), and the mean field (c); cuts are given in the SM.

Three regions appear. (i) For $U < U_c(g)$ the loop is a 0 loop: the singlet is the ground state, the response is diamagnetic and the mode is stiffened, $\omega' = \omega\sqrt{1 + 2g^2E''(0)/\hbar\omega}$ (up to 9.4ω at $g = 0.4$), with the squeezed vacuum of the stiffened mode as ground state (its photon number reproduces the DMRG value, 1.89 against 1.91 at $g = 0.4$); this is the inductive shift that reads out the parity of dot junctions [29, 30]. The boundary is the adiabatic end of Fig. 2: it moves from $U_c^0 = 0.94\Delta$ to 0.85Δ at $g = 0.4$, the crossing of the *global* minima $E_{\mathrm{odd}}(\pi/2) = E_{\mathrm{even}}(0)$; the window between is the π' phase of the junction [23, 24], which a loop with a free flux leaves by letting $\Phi_0/2$ in. (ii) For $U > U_c$ and $g > g_c(U)$ the mode condenses: at $U = 2\Delta$ the photon number rises from 0.5 at $g = 0.11$ to 7.3 at $g = 0.14$, where the current reaches $0.96I_c$, and then falls as $I \simeq (e/\hbar)\hbar\omega\varphi^\star/2g^2$ because the flux saturates at $\Phi_0/2$ while the inductance $\propto g^2/\hbar\omega$ grows. The threshold ($|\langle I\rangle| = 0.5I_c$) lies *above* $\beta = 1$ by $\Delta g = 0.02$, 0.04 and 0.12 at $U = 2, 4, 6\Delta$ ($E_J/\hbar\omega = 10$, 5, 3.2): the zero-point motion of the flux opposes the condensation, and the crossover broadens from one grid step to $\Delta g \simeq 0.08$. (iii) At $U = 8\Delta$ ($E_J/\hbar\omega = 2.1$) no ordered phase is left for $g \le 0.4$ although β reaches 2.7: the ground state is the cat state with $\delta \simeq \hbar\omega$ and a current $\le 0.04I_c$ proportional to the seed. The seed-free map (b) has the same boundary as the contour $\delta = 0.1\hbar\omega$ and shows the exponential collapse of δ inside the ordered region.

Above threshold the full DMRG and the circuit model agree to 0.2% of I_c on average (2.4% at worst), to 0.1 in the photon number and to $10^{-5}\Delta$ in energy: at $\hbar\omega/E_A \sim 10^{-3}$ Eq. (2) with the microscopic $E_s(\varphi)$ is the exact circuit model. Mean field locates g_c to within the grid step for $U \le 5\Delta$ but underestimates the current by 10–20% (the product state renormalizes the junction bond by $|\langle D\rangle| < 1$), overestimates the photon number at large g (5.2 against 3.7 at $U = 2\Delta$, $g = 0.4$), predicts an ordered phase with $0.6I_c$ at $U = 7$–8Δ that does not exist, and, with its Hartree–Fock boundary $U_c^{\mathrm{MF}} = 0.82\Delta$, orders the loop at $U = 0.85\Delta$ with $0.95I_c$ where the exact ground state is the singlet 0 loop. Ref. [20] applies this ansatz to the same circuit; the exact solution limits its validity to $E_J \gg \hbar\omega$ and U well above U_c, and for $E_J \simeq \hbar\omega$ the time-reversal-breaking state it predicts is absent (SM).

Cat state and macroscopic tunnelling. Without a seed H is real, $\langle I\rangle = 0$ in every eigenstate, and the order is carried by the two lowest states of the doublet sector [Fig. 4, $U = 2\Delta$, $E_J/\hbar\omega = 10$]: their splitting δ collapses and the flux projected on them acquires the eigenvalues $\pm\Phi^\star$. DMRG and the circuit model give the same $\Phi^\star$ to 0.2% ($0.59\,\Phi_0/2$ at $\beta = 2$), the same photon number to 0.03, and $\delta/\hbar\omega = 0.70$, 0.24, 2.65×10^{-4} at $\beta = 0.5$, 1, 1.5; at $\beta = 1.5$ the DMRG values at $\chi = 300$ and 400 are 2.66 and 2.65×10^{-4}, so the macroscopic tunnelling of a condensate of seven photons ($\delta = 2\times10^{-7}\Delta$) is resolved directly. Deeper in the ordered phase δ falls below the DMRG resolution for excited states ($4\times10^{-5}\hbar\omega$) and the circuit model, diagonalized in the two flux-parity sectors separately, gives 8×10^{-7} and 6.5×10^{-9} at $\beta = 2$ and 3 (SM). The DMRG ground state is the cat state itself: $|\langle\Phi\rangle| < 10^{-3}\Phi^\star$, $P(\Phi)$ is bimodal with peaks at $\pm\Phi^\star$ [Fig. 4(c)], and a DMRG started from the polarized coherent state relaxes to the same symmetric state to $10^{-10}\Delta$. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [Fig. 4(b)].

The two current directions are related by time reversal and δ is the only scale that separates them. In the spin-degenerate model each is also Kramers degenerate in the spin of the dot, giving the manifold $\{\uparrow,\downarrow\}\otimes\{\circlearrowright,\circlearrowleft\}$; with spin–orbit coupling, as in InAs or Ge, the doublet surfaces are spin dependent [30, 33, 47], the wells are spin locked and the manifold is one Kramers pair: the loop is then the inductively protected spin qubit of Ref. [32], of which Eq. (1) with a spin-dependent t_J is the microscopic model. A flux offset $\delta\Phi > \delta/2I^\star$ polarizes the loop in any realization; the intrinsic π junction removes the need for a $\Phi_0/2$ bias, not the sensitivity to it. A weak link biased by $\Phi_0/2$ gives the same ordered state, with its direction

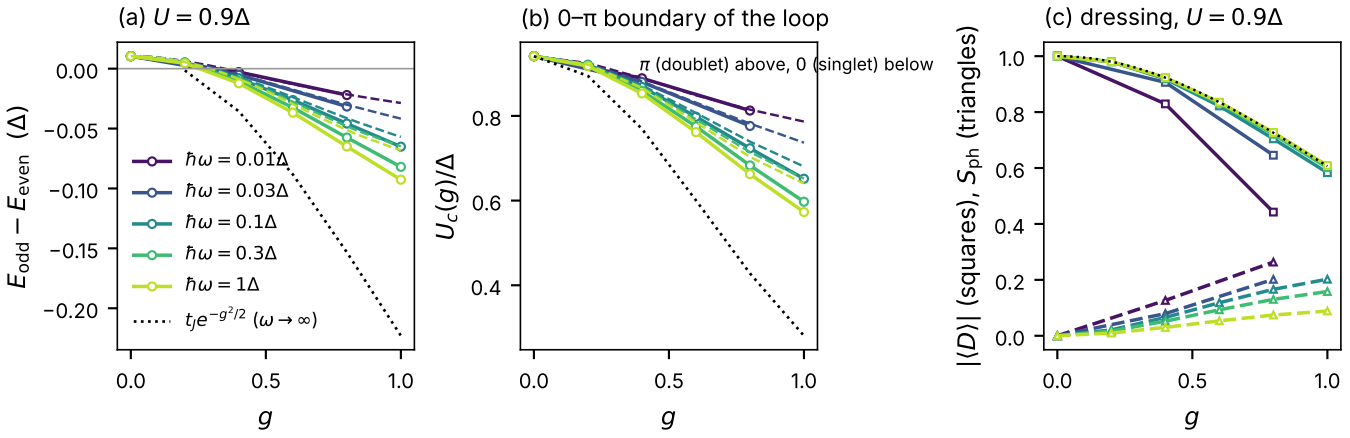


FIG. 2. Zero-point flux and the 0–π transition ($L = 10$, $t_J = 0.5\Delta$, $\varphi_{cl} = 0$, no seed). (a) Doublet–singlet energy difference versus g at $U = 0.9\Delta$ for $\hbar\omega = 0.01$–$1\,\Delta$: full DMRG (symbols), circuit model (2) with the $g = 0$ surfaces (dashed), Lang–Firsov ring with $t_J e^{-g^2/2}$ (dotted, the $\omega \to \infty$ limit). The sign change is the switch of the loop from 0 to π. (b) Boundary $U_c(g, \omega)$ of the loop from the crossing $E_{odd} = E_{even}$ (same symbols and lines). (c) Dressing of the junction in the doublet sector at $U = 0.9\Delta$: $|\langle D\rangle|$ (squares; dotted: $e^{-g^2/2}$) and the photon–electron entanglement entropy S_{ph} (triangles).

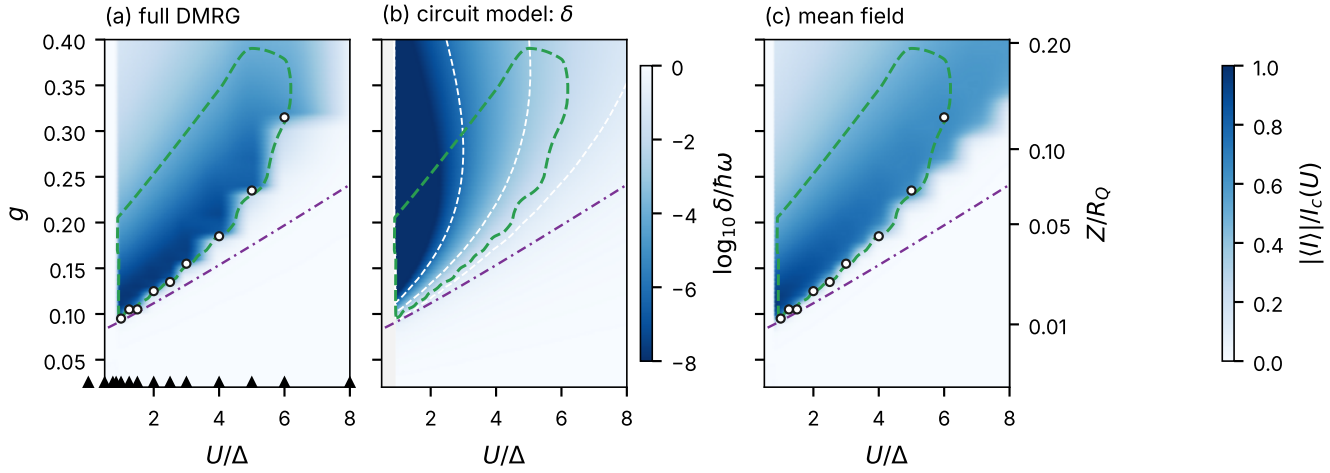


FIG. 3. Phase diagram of the loop in the plane of U/Δ and g ($Z/R_Q = 4g^2/\pi$, right axis) at $\hbar\omega = 7.39 \times 10^{-4}\Delta$, $\varphi_{cl} = 0$. (a) Persistent current $|\langle I\rangle|/I_c(U)$ from the full DMRG with seed 10^{-3} (triangles: computed U; circles: $|\langle I\rangle| = 0.5 I_c$). (b) Seed-free order parameter of the circuit model (2): tunnel splitting δ between the two current directions (white contours $\delta/\hbar\omega = 10^{-1}, 10^{-3}, 10^{-6}$; grey: singlet 0 loop). (c) Mean field. Green dashed: $0.5 I_c$ contour of the circuit model; purple dash-dotted: $\beta = 1$.

selected by the residual h/e harmonic (SM).

Where to look. The flux-driven 0–π switch of Fig. 2 needs $\hbar\omega = 0.03$–0.3Δ and $g = 0.3$–0.6: for Al ($\Delta = 180\,\mu$eV) a mode at 1–13 GHz sharing its inductance with the loop and an impedance $Z = 4R_Q g^2/\pi = 0.7$–$3\,\mathrm{k}\Omega$, the range of granular-Al or junction-array superinductors [42, 48], with the dot gate tuned to within $\sim 0.1\Delta$ of its singlet–doublet boundary as in Refs. [29, 31]. It is read out as the jump of the resonator frequency between the stiffened 0 phase and the softened π phase, and its signature is that it is driven by the impedance (the superinductor length) at fixed gate voltage.

The persistent-current phase and the cat state require $E_J/\hbar\omega \gtrsim 5$ and $\hbar\omega > k_B T$, which is restrictive: the junction of the phase diagram ($U = 2\Delta$, $\Gamma = \Delta/2$, $E_J = 7.4\times10^{-3}\Delta$) has $E_J/h = 0.3\,$GHz and $I_c = 0.65\,$nA in Al, so $\hbar\omega = E_J/10$ would be 32 MHz, far below the electron temperature. A stronger junction ($\Gamma = 2\Delta$, $U = 6\Delta$, $E_J = 0.034\Delta$, SM) in NbTiN or Nb ($\Delta \simeq 1.2$–$1.4\,$meV) gives $E_J/h \simeq 10\,$GHz, $I_c \simeq 20\,$nA, a $\simeq 1\,$GHz mode (50 mK) at $E_J/\hbar\omega = 10$, $L \simeq 17\,$nH and $Z \simeq 160\,\Omega$ at $\beta = 1$: a π-rf-SQUID at zero applied flux, for which the full DMRG gives a condensate of 9.6 photons and $0.94 I_c$ (SM). A nA dot junction with a GHz superinductor has

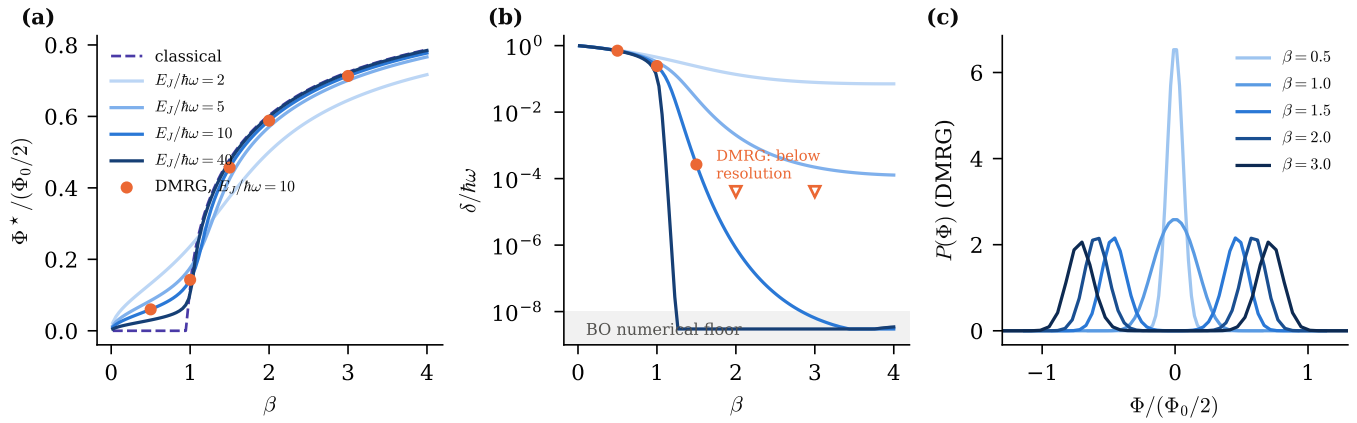


FIG. 4. Cat state of the π loop ($U = 2\Delta$, $L = 10$, $\hbar\omega = E_J/10$, no flux, no seed). (a) Condensate flux $\Phi^\star$ versus β for increasing $E_J/\hbar\omega$ (lines: circuit model with the DMRG $E(\varphi)$; circles: full DMRG at $E_J/\hbar\omega = 10$; dashed: classical). (b) Tunnel splitting (triangles: DMRG below its resolution). (c) Flux distribution of the DMRG ground state.

$E_J/\hbar\omega \lesssim 3$ and no current-carrying phase for $g \leq 0.4$ (the $U = 8\Delta$ line of Fig. 3): its ground state is the cat state with $\delta \sim \hbar\omega$, visible as a dip of the $|0\rangle \to |1\rangle$ frequency and a jump of $\langle 0|\Phi|1\rangle$ in two-tone spectroscopy, where mean field predicts a current of $0.6I_c$. SFS junctions ($E_J/\hbar\omega \sim 10^2$–$10^4$) are the classical limit [3, 4].

Conclusion. Treating the electrons of a quantum-dot Josephson junction and the quantized flux of its loop on the same footing, we find that the vacuum of the mode controls the junction: its zero-point flux converts a 0 junction into a π junction, adiabatically through the curvatures of the singlet and doublet surfaces and non-adiabatically through the Franck–Condon reduction of the level width. Deep in the adiabatic regime the loop is exactly a fluxonium with the microscopic $E(\varphi)$, the "photon condensate" is the screening instability of a π loop, and the exact ground state is the cat state of the two current directions; the product-state mean field fails at small $E_J/\hbar\omega$ and at the 0–π boundary. The method extends to Andreev spin qubits with spin–orbit coupling, to multi-dot junctions, and to a mode coupled to a transmission line.

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Supplemental Material

S1. Gauge covariance of the static phase and the circuit meaning of the one-bond vertex. For a static phase the position of the Peierls factor is a gauge choice: with the pairing transformed covariantly, $\Delta_j \to \Delta e^{-2i\theta_j}$, a uniform vector potential along the ring gives the same BdG spectrum as the full phase on the junction bond (prescription of Ref. [38]; verified to 10^{-10} for $\varphi = 0$–$\pi/2$; a uniform vector potential with a *real* Δ is not equivalent). For the quantized flux the unitary $W = \exp[-ig(a+a^\dagger)\sum_j \lambda_j n_j]$ that moves a fraction λ_j of the phase onto bond j does not commute with $\hbar\omega a^\dagger a$ [39]: with $N = \sum_j \lambda_j n_j$ and $P = -i(a-a^\dagger)$, $W^\dagger a^\dagger a W = a^\dagger a + gNP + g^2N^2$, so that the two placements differ by $\hbar\omega g(NP+gN^2)$, a coupling of the electron charge on the displaced sites to the charge of the mode. In a lumped circuit this term is the capacitive coupling of the mode to the ring segment that carries the vector potential; Eq. (1) of the main text is the circuit in which the voltage of the mode drops across the junction, the only high-impedance element of the loop. The neglected terms are of relative order $\hbar\omega g^2/E_A$: 10^{-3} on the phase diagram, 10^{-2}–10^{-1} at the largest $\hbar\omega$ and g of Fig. 2, where the flux-driven switch is already fully developed at $\hbar\omega = 0.01\Delta$.

S2. Junction at $g = 0$ and the π' window. Figure S1 characterizes the junction alone. DMRG in the even and odd parity sectors gives the 0–π transition of the Anderson impurity between BCS leads [23–25, 49]. At $U = 2\Delta$, $E_2 = +7.39 \times 10^{-3}\Delta$ with $|E_1|/E_2 = 2\%$ and $|E_4|/E_2 < 1\%$: a pure π junction, indistinguishable after normalization from a weak link biased by $\Phi_0/2$ [Fig. S1(a)]. E_2 decreases monotonically with U ($\propto U^{-1.3}$ for $4\Delta \le U \le 8\Delta$, approaching the $1/U$ of cotunnelling through the singly occupied level). Three crossings of the two sectors are distinct [Fig. S1(b)]: the zero-flux crossing $U_c^0 = 0.94\Delta$, the crossing of the global minima $E_{\rm odd}(\pi/2) = E_{\rm even}(0)$ at 0.85Δ, and the crossing at half a flux quantum below it. The window $0.85\Delta < U < 0.94\Delta$ is the π' phase [23, 24]: the singlet is the ground state at zero flux, but the global minimum of $E(\varphi)$ belongs to the doublet at $\Phi_0/2$. The singlet surface has a cusp at $\varphi = \pm\pi/2$, where two even-parity states cross, and is represented by a spline through 25 phases in the circuit model; the doublet surface is a smooth $\cos 2\varphi$ with 2% corrections. The strong junction used in the feasibility estimate ($t_J = t$, $\Gamma = 2\Delta$) has $U_c^0 = 5.55\Delta$ and, at $U = 6\Delta$, $E_2 = 0.034\Delta$, $E_{\rm odd} - E_{\rm even} = -0.044\Delta$ at zero flux and $|E_1|/E_2 = 1\%$ ($L = 14$); its full-DMRG scan at $\hbar\omega = 0.003t$ gives a condensate of 9.6 photons and $0.94I_c$ at $g \simeq 0.12$, in agreement with the circuit model to 1%.

S3. DMRG protocol. The ring is stored in folded order next to the bosonic site, so that all ring bonds have range ≤ 2 and the vertex $Dc_{1\sigma}^\dagger c_{0\sigma}$ is a three-site operator. Fermion parity and S^z are conserved. Benchmarks: at $g = 0$ the energies of the non-interacting ring agree with exact BdG to 10^{-14} at $\chi = 1024$; for a four-site ring with a photon, DMRG reproduces exact diagonalization and the sum rule $\langle I\rangle = (e/\hbar)(\hbar\omega/2g)\langle X\rangle$ to 10^{-7}. On the phase diagram $\chi = 300$ (truncation error $< 2\times10^{-8}$; $\chi = 400$ changes the energies by $< 4\times10^{-8}\Delta$ and the current by $< 0.3\%$ away from the crossover points) and $N_{\rm ph} = 50$–62 (last Fock weight $< 10^{-7}$; $N_{\rm ph} = 80$ changes the energy by $< 10^{-9}\Delta$). The relevant criterion is not the energy: the two current states are split only by δ and by the seed, $2I^\star\varphi_{\rm cl} \sim 10^{-5}\Delta$, and the local updates of DMRG drift towards the polarized state over tens of sweeps and can stop in an under-polarized state whose energy is within $10^{-6}\Delta$ of the ground state. Every seeded point is therefore started from two independent states, a warm start from the neighbouring point on the vacuum side and a product state with the mode in a coherent state at the classical flux, converged first with seeds 10^{-2} and 3×10^{-3} and then relaxed at 10^{-3}; the lower energy is kept when the two differ by more than $3\times10^{-7}\Delta$, otherwise the pair is reported as a bracket (47 of 427 points, all inside the crossover windows); a state is accepted only if it satisfies the sum rule to $10^{-3}I_c$ (median $10^{-4}I_c$); and the current above threshold must be independent of the seed (seed 10^{-4}: unchanged to 0.4% above threshold, reduced by the factor 0.09 below) [Fig. S2]. An earlier version of these scans evaluated the observables after a warm start with the displacement operator of the previous g; the sum rule exposed this error. For the seed-free runs (cat state, flux-driven switch) the symmetric state is the ground state in each sector and no such care is needed; on the grid of Fig. 2 ($\chi = 200$, $N_{\rm ph} = 20$–48) the truncation error is $< 4\times10^{-7}$ and the last Fock weight $< 4\times10^{-8}$, and the boundary U_c is obtained by linear interpolation of $E_{\rm odd} - E_{\rm even}$ between U nodes 0.1–0.2Δ apart (the difference is linear in U to 10^{-2}); the photon–electron entanglement entropy is computed from the photon reduced density matrix. The circuit model is diagonalized in a Fock space of 200 states (DVR in the eigenbasis of X); for the tunnel splitting below $10^{-6}\hbar\omega$ the two flux-parity blocks are diagonalized separately, which gives δ as a difference of two independently converged energies ($6.46\times10^{-9}\hbar\omega$ against 6.51×10^{-9} from the full matrix at $\beta = 3$, $E_J/\hbar\omega = 10$; at $E_J/\hbar\omega = 40$ both are at the double-precision floor $\sim 10^{-9}\hbar\omega$ for $\beta \ge 1.25$).

S4. Half-flux-biased weak link. For an ordinary weak link ($L = 14$, $t_J = 0.5$, $E_2 = -0.056\Delta$, $E_1 = -5.4\times10^{-4}\Delta$, Andreev gap 0.77Δ) biased by $\Phi_0/2$, Fig. S4 shows the full DMRG at $E_J/\hbar\omega = 10$ versus β. The current rises at $\beta \simeq 1.1$, saturates at $0.96I_c$ for $\beta \approx 1.7$ and then decreases as the flux approaches $\Phi_0/2$. DMRG and the circuit model agree to $2\times10^{-3}\hbar\omega$ in energy, 2% of I_c in current and 1.3% in $E_1 - E_0$. The product state

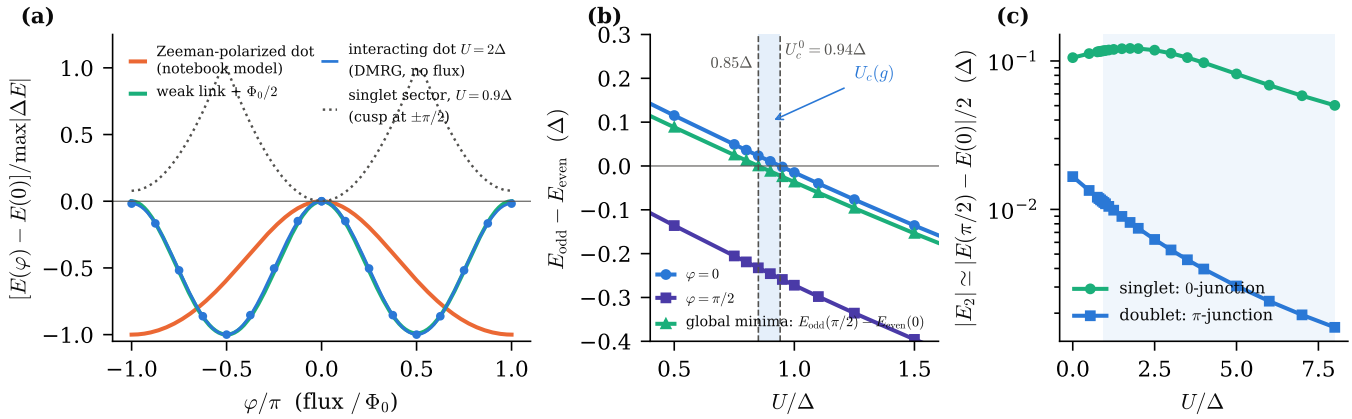


FIG. S1. (a) Normalized $E(\varphi)$ of the loop for a Zeeman-polarized dot in a short ring (dominated by the h/e harmonic), a weak link biased by $\Phi_0/2$ ($L = 14$) and the interacting dot at $U = 2\Delta$ without bias ($L = 10$; symbols: DMRG; line: cosine fit); dotted: the singlet surface at $U = 0.9\Delta$ with its cusp. (b) $E_{\rm odd} - E_{\rm even}$ at $\varphi = 0$, at $\varphi = \pi/2$ and between the global minima (DMRG, $L = 10$, $t_J = 0.5\Delta$); shaded: the π' window in which the boundary of the loop moves with g. (c) $|E_2|$ in the singlet and doublet sectors.

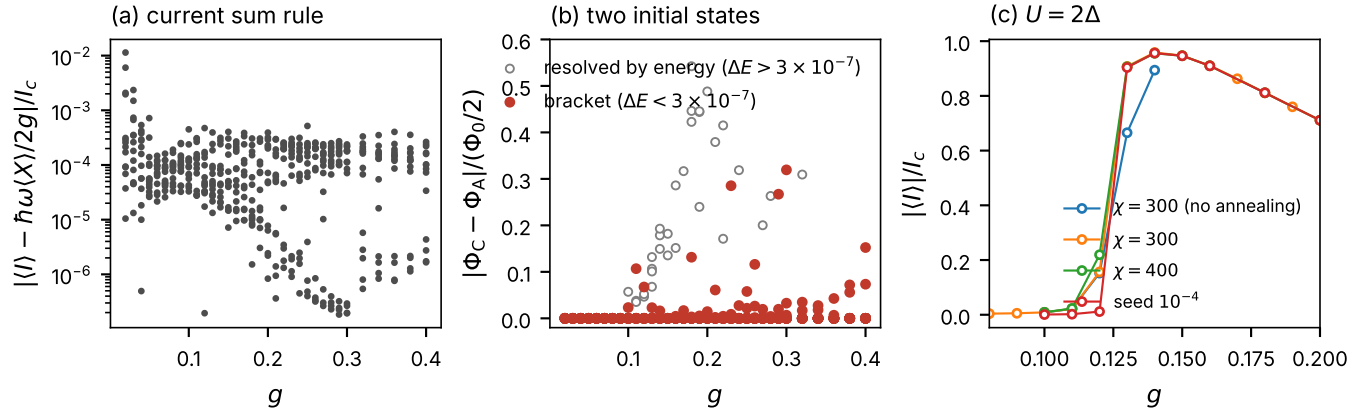


FIG. S2. Convergence of the full DMRG on the phase diagram. (a) Violation of the sum rule over the grid. (b) Difference between the fluxes of the two initial states (vacuum side A, polarized side C) in units of $\Phi_0/2$: open symbols resolved by the energy, filled reported as brackets. (c) $U = 2\Delta$ near threshold: $\chi = 300$ without and with the annealed start, $\chi = 400$, and seed 10^{-4}.

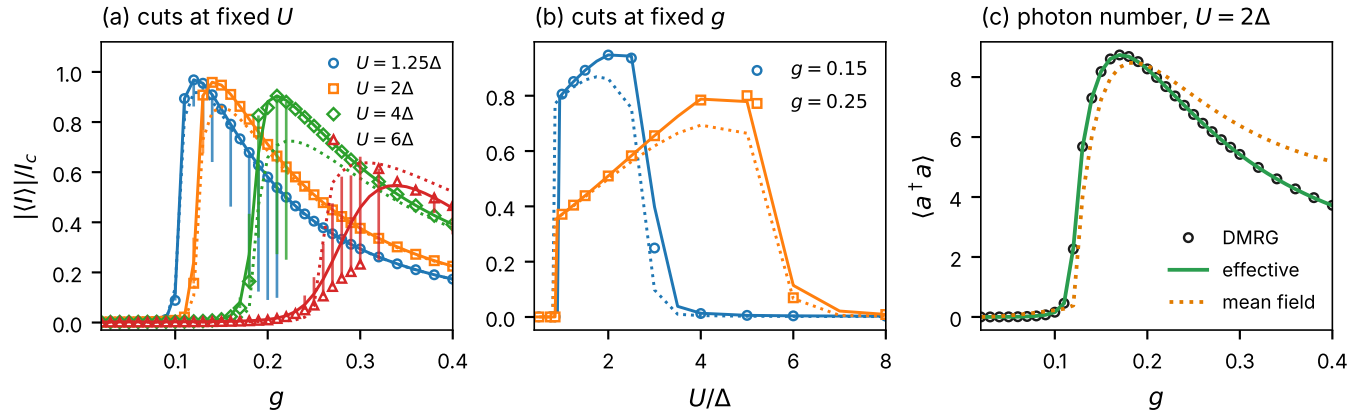


FIG. S3. Cuts through Fig. 2 of the main text. (a) Current versus g at $U/\Delta = 1.25, 2, 4, 6$ and (b) versus U at $g = 0.15, 0.25$: full DMRG (symbols; bars: bracket between the two initial states), circuit model (solid), mean field (dotted). (c) Photon number at $U = 2\Delta$.

is 0.2–0.4$\hbar\omega$ too high in energy and overestimates the current threefold near threshold ($0.71I_c$ against $0.22I_c$ at $\beta = 1.13$). The direction of the current is selected by the residual h/e harmonic, which tilts the double well by $2|E_1|\sin\varphi^\star \approx 0.2\hbar\omega$ and sets $E_1 - E_0$ for $\beta > 1.2$; without it $\delta/\hbar\omega$ would fall below 10^{-6} at $\beta = 2$. The flux distribution evolves from a Gaussian at zero to a single displaced peak.

S5. Product state at $E_J \simeq \hbar\omega$. The product-state mean field is the method of Refs. [17, 20] and of our own earlier treatment of this circuit: Hartree–Fock BdG electrons with the junction bond renormalized by $\alpha = \langle D\rangle$, times the ground state of $\hbar\omega a^\dagger a - (\eta D + \text{h.c.})$ with $\eta = t_J e^{i\varphi}\langle c_1^\dagger c_0\rangle$, iterated to self-consistency. For a Zeeman-polarized dot in a short BCS ring ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$; energies in units of t) it predicts a current-carrying phase for $g \gtrsim 0.75$ with $|\langle I\rangle| \approx 0.02$ and $\langle a^\dagger a\rangle \simeq 1$–$1.4$. DMRG shows that this phase is an artefact [Fig. S5]: the exact energy lies below the MF energy by up to $1.8\hbar\omega$, the gap to the first excited state stays at $0.33\hbar\omega$, the photon number saturates at 0.36 and the response to the seed is linear, $\langle I\rangle \approx 7\times10^{-5}$. Two effects are responsible. (i) The product state is not gauge covariant: with the full Peierls phase on one bond it renormalizes that bond by $|\langle D\rangle| < 1$ and loses phase-*independent* kinetic energy, which the exact state recovers by following the flux. (ii) $E_J \simeq \hbar\omega$: the Landau energy develops a double well for $g > 0.58$, but its depth is below the zero-point energy and $P(X)$ stays unimodal. In this ring $L \lesssim \xi$ and $E(\varphi)$ is dominated by the h/e harmonic ($E_1 = 2.1\times10^{-2} \gg E_2 = 2\times10^{-3}$), which is why all other results use $L \gg \xi$. At $\hbar\omega = 0.002$ the same model does order, at $g_c \simeq 0.21$, while mean field gives 0.30.

S6. Physical units. With $E_J/h = 0.5\,\text{GHz}\times I_c[\text{nA}]$ and $\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$:

junction, material	E_J/h	I_c	$E_J/10h$	$E_J/10k_B$	$L_{\beta=1}$
2Δ, $\Delta/2$, Al	0.32 GHz	0.65 nA	32 MHz	1.5 mK	510 nH
2Δ, $\Delta/2$, NbTiN	2.1 GHz	4.3 nA	210 MHz	10 mK	76 nH
6Δ, 2Δ, Al	1.5 GHz	3.0 nA	150 MHz	7 mK	110 nH
6Δ, 2Δ, NbTiN	9.9 GHz	20 nA	0.99 GHz	47 mK	17 nH
6Δ, 2Δ, Nb	11.5 GHz	23 nA	1.15 GHz	55 mK	14 nH

(first column: U, Γ, superconductor; $L_{\beta=1} = \Phi_0/2\pi I_c$). The flux-driven switch needs none of this: $\hbar\omega = 0.1\Delta$ is 4.4 GHz in Al and $g = 0.5$ is $Z = 2.0\,\text{k}\Omega$.

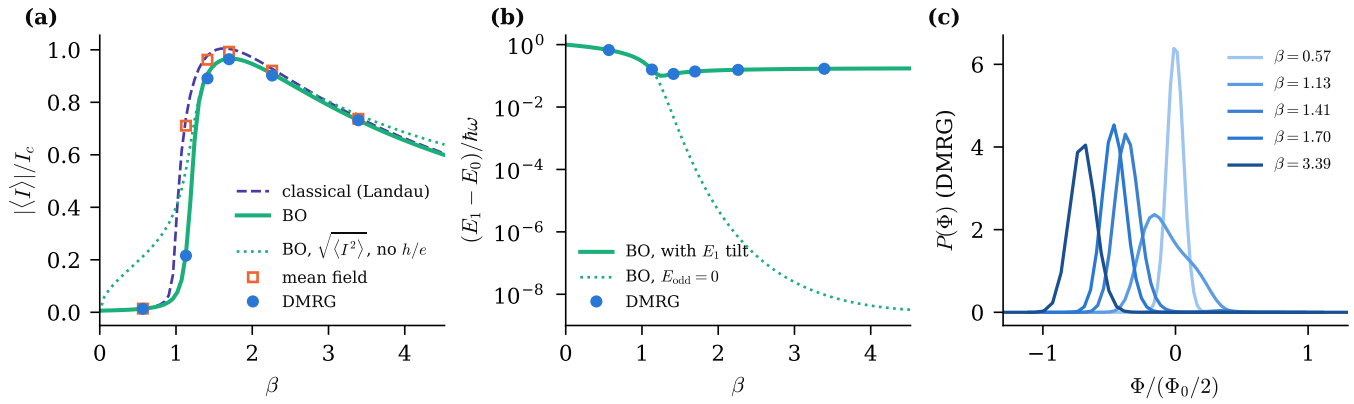


FIG. S4. Weak-link loop ($L = 14$) biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current versus β: DMRG (circles), MF (squares), circuit model with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state.

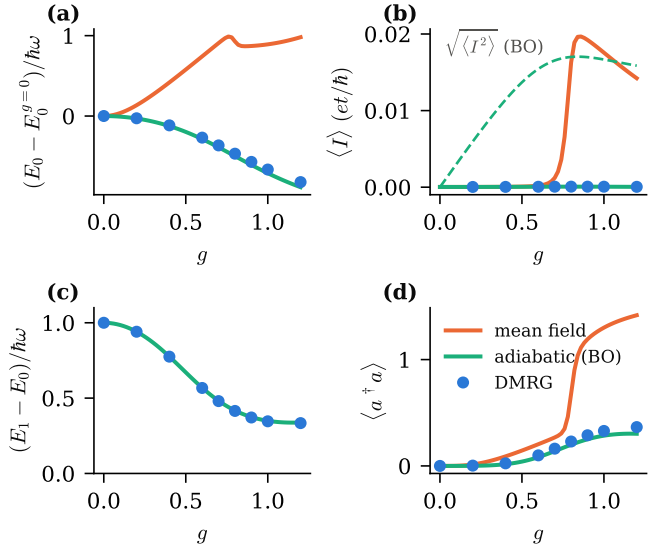


FIG. S5. Short ring with a Zeeman-polarized dot ($\hbar\omega = 0.02$, seed 10^{-3}): mean field (orange), circuit model (green) and DMRG (symbols). (a) Energy, (b) current (dashed: rms current of the circuit-model ground state), (c) gap to the first excited state, (d) photon number.